MIL-W-13741(Ord)
3 November 1954
SUPERSEDING
Army 55-5A
28 October 1926

MILITARY SPECIFICATION

WATCH, POCKET, GRADE II

1. SCOPE

1.1 This specification covers a pocket watch for use by the Armed Forces for general purposes.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and drawings, and specifications and drawings referenced thereon, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

TT-R-58	Radioactive Luminous Compound and Adhesives

Military

MIL-P-11446	Packaging of Watches, Pocket, Stop and Wrist

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Ordnance Corps

B7197860	Shock Absorber
A7197861	Chain, Leather

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 The Grade II pocket watch shall be equal to the sample furnished by the Government or equal to the sample furnished by the contractor if same has been approved by the Government.

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3.1.1 Each bidder shall submit with his proposal, charges prepaid, two sample watches, also two cases without movements, of the type he proposes to furnish. He shall also submit with his proposal the characteristics and full description of the movement which he proposes to furnish. The type of movement, quality, and workmanship shall be considered in awarding the contract. The samples furnished by the successful bidder will be held for the guidance of the inspector without cost to the Government and will be returned upon request, unless needed for further reference.

3.2 Material.- Where a definite material is not specified it shall be of a composition and quality that is suitable for watches which must meet the other requirements of this specification. All materials shall be of uniform quality and free of any defects which might impair the functioning of the watch.

3.3 Characteristics.- The pocket watch shall have the following characteristics:

3.3.1 Movement shall be preferably of standard American manufacture, nickel, 3/4 plate or bridge, 16 size, with stem winding, pendant setting, double-roller escapement, split or compensation balance, safety center pinion, Breguet hairspring, micrometric or patent regulator, as ordered.

3.3.2 Where a jeweled movement is being furnished, the balance, plate, cap, and hole jewels shall be set in individual mountings and shall be secured in a plate or bridge by means of screws or a forced fit, preferably by screws.

3.3.3 Entrance for the stem shall be located at a point above the figure 12 on the dial.

3.3.4 Face of dial shall be of white porcelain, full face, single or double sunk.

3.3.5 Dial shall be of the Montgomery safety numerical dial type.

3.3.6 Figures and marking shall be an integral part of the basic porcelain and shall be black, with the exception of the numerals indicating minute periods, which shall be in red.

3.3.7 Dial shall be manufactured with the figures flush with surface of the porcelain or slightly depressed.

3.3.8 The hour and minute hands shall be of blued steel, heavy spade type, distinctly different in length, and shall have a channel-shaped center for receiving a medium-thick coating of permanent adhesive luminous material conforming to Specification TT-R-58 applied in such a manner that it will not loosen readily or fall from the hands. This material may be omitted from the second hand. The numerals indicating hours shall receive the luminous

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coating, which shall be applied in such a manner as to give lighted numerals without a halo. The red numerals indicating the minute periods and numerals indicating seconds shall not receive the luminous coating.

3.3.9 The second hand shall be made of steel.

3.3.10 Case shall be of nickel or an approved nickel alloy. No nickel-plated alloy shall be permitted. It shall be round, with antique bow, inner cap or snap-back hinged, and with snap or hinged bezel, preferably hinged.

3.3.11 The winding and setting crown shall be constructed with a snug fit between the sleeve and stem, in order to prevent their becoming loose or being turned accidentally.

3.3.12 The assembled case shall be as nearly dust and moisture proof as possible.

3.3.13 The outside surface of the metal case shall have a bright nickel finish.

3.3.14 The bezel shall be fitted with an unbreakable crystal made of Columbia Resin (CR-39) or substantial equal.

3.3.15 Luminous material shall be Grade 65M, in accordance with Specification TT-R-58, using Type A adhesive.

3.3.16 Lubricant used in the movement shall be approved by the contracting agency. Bidders shall state with their bids the type of lubricant they intend to furnish, together with a complete description of the physical and chemical characteristics of the lubricant.

3.4 Interchangeability.- Like parts shall be interchangeable in all like watches supplied by one manufacturer.

3.5 Marking.- Special marking accuracy shall be required as follows:

(a) Dial markings uniform, plain and easily distinguishable.

(b) Spacings of markings, width and length of lines.

3.6 Shock absorber and leather chain.

3.6.1 The shock absorber shall be in accordance with Drawing B7197860 or equal to sample furnished by the contractor if same has been approved by the Government.

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3.6.2 The leather chain shall be in accordance with Drawing A7197861 or equal to sample furnished by the contractor if same has been approved by the Government.

3.7 Accuracy.- The watch shall meet all accuracy requirements specified herein.

3.7.1 Watch accuracy requirements at extreme temperatures shall be tested in a dial up position for the periods specified in 4.3.1.7.

3.7.2 The movement shall be adjusted to isochronism.

3.8 Identification.- Each watch shall show on the movement (barrel bridge or train bridge or both) the manufacturer's name and serial number and manufacturer's model or grade number. Each case shall be legibly marked on the exterior of the back with the Ordnance grade number below the words "U. S. Government".

3.9 Workmanship.- Standards of workmanship shall be such that watches will meet the requirements of 3.1 through 3.8. All components, sub-assemblies, parts, and the final assembly shall meet the inspection requirements for defects for quality and dimensions as specified in 4.2. through 4.2.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling.- Samples for conducting the inspection specified in 4.2 and tests specified in 4.3, shall be selected in accordance with Table III, Inspection Level II, and Table IV-B, AQL 2.5 of Standard MIL-STD-105.

4.2 Inspection

4.2.1 Each sample shall be inspected to determine the following:

(a) Winding stem and setting-sleeve are carefully adjusted so that the winding stem lines up with the movement to avoid any possible friction causing hard winding. Stem is set close to the case.

(b) Hinges and snap parts of the case operate properly and parts fit snugly.

4.3 Tests

4.3.1 Accuracy.- The movement shall be tested in three positions. These tests shall be for six periods of 3 days each. During these tests a chronometer shall be used for checking purposes.

4.3.1.1 One test shall be in the vertical position with the figure 12 of the dial up, one in a horizontal position with the dial up, and one in a horizontal position with the dial down.

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4.3.1.2 The first two periods shall be in a vertical position with the pendant up, the second two periods shall be in a horizontal position with the dial up, and the last two periods shall be in horizontal position with the dial down.

4.3.1.3 The total time gained or lost in any of the six periods must not exceed 90 seconds.

4.3.1.4 The difference in the time gained or lost between periods one and two (both vertical) and three and four (both horizontal) shall not exceed 30 seconds.

4.3.1.5 The difference in the time gained or lost between periods one and two (both vertical) and five and six (both horizontal) shall not exceed 30 seconds.

4.3.1.6 The difference in the time gained or lost between periods two and three (change of position) shall not exceed 75 seconds. Watches shall be wound and readings taken daily.

4.3.1.7 In addition to the position tests, the movement shall also be subjected to temperature tests for two periods of 3 days each at 40° and 90°F.

4.3.1.8 The range of the mean daily rates for the two temperature tests shall not exceed 30 seconds.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Specification MIL-P-11446.

6. NOTES

6.1 Ordering data.- Invitation for bids and contracts should specify the title, number, date of this specification, and type of packing.

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